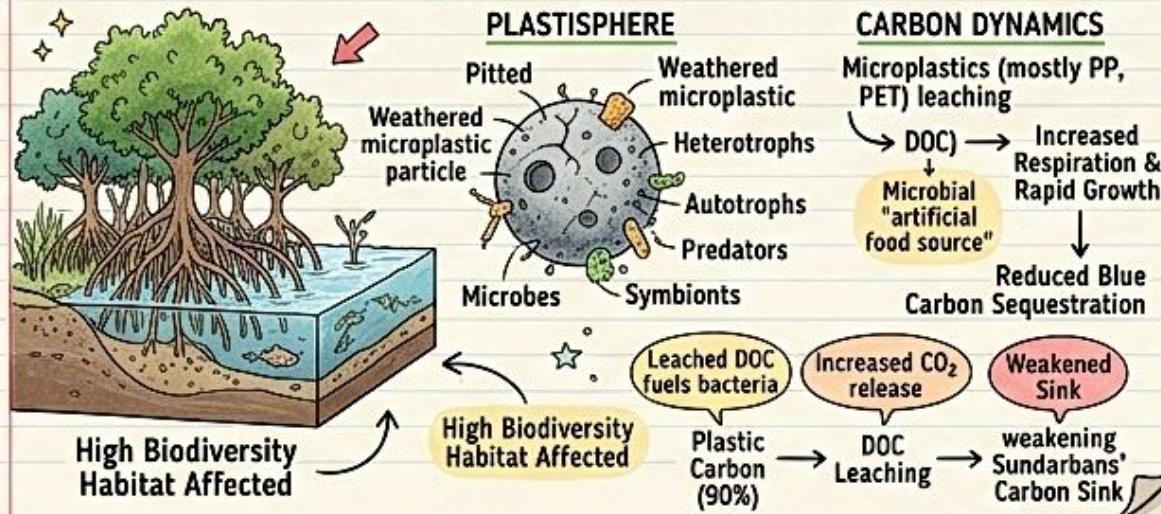
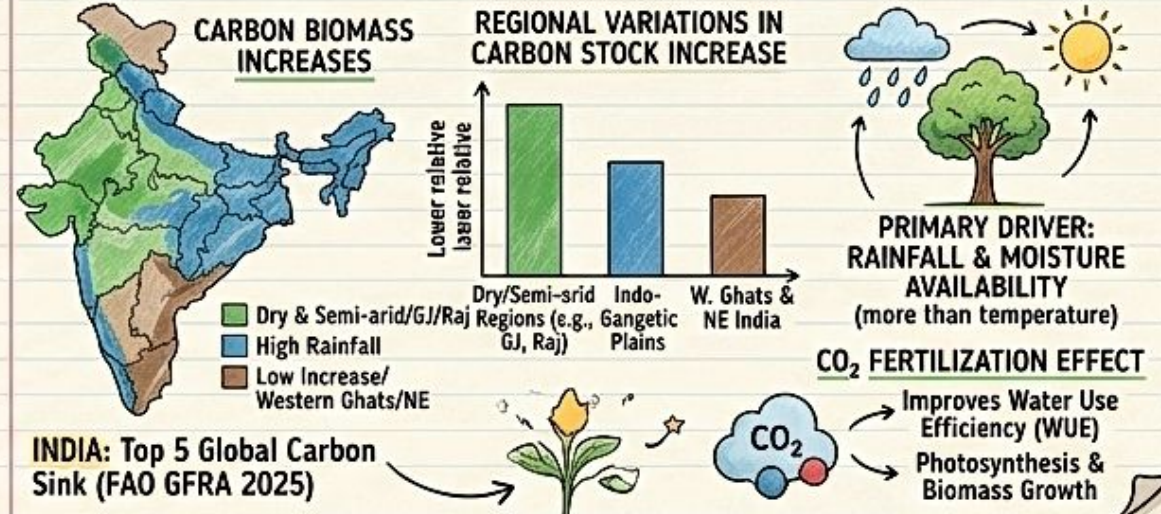


UPSC CSE PRELIMS 2026: KEY TOPICS

1 SUNDARBANS: MICROPLASTICS ALTERING CARBON DYNAMICS



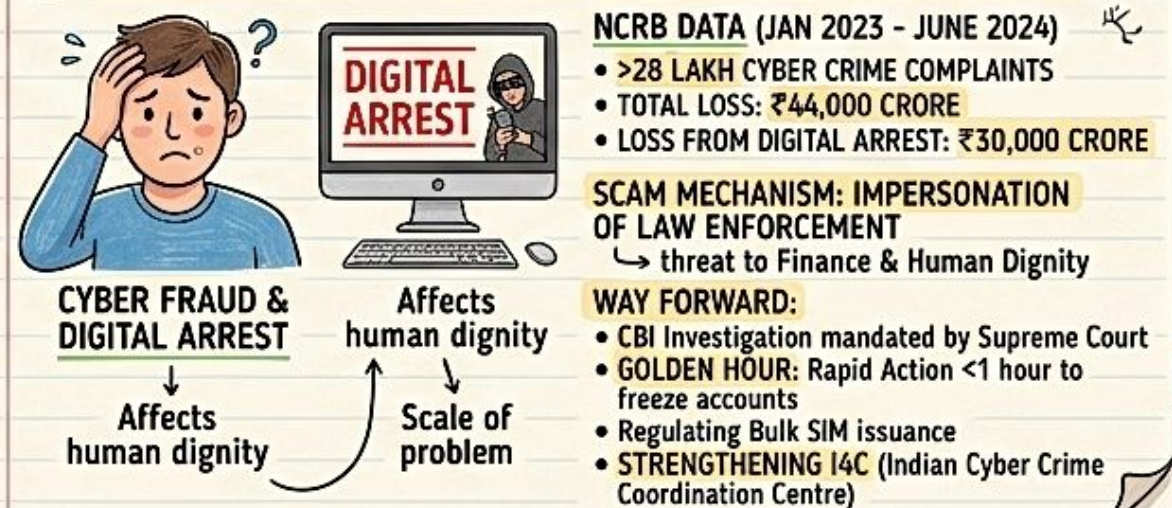
2 INDIA'S CARBON SINK: FOREST CARBON BIOMASS TRENDS



3 INDIA & KOREA CEPA: DEEPENING COOPERATION



4 CJI CONCERNS: CYBER CRIME & DIGITAL ARREST SCAM



1. SEBI & SOCIAL STOCK EXCHANGE (SSE) AMENDMENTS

SOCIAL IMPACT FUNDS (SIF):

- Cat I AIF
- Invests in NPOs/FPEs for measurable impact.

RETAIL PARTICIPATION



Minimum Investment Threshold (Individuals) → Slashed from ₹2 Lakh to ₹1,000

↓
Matches ZCZP minimum application.

ZCZP BONDS
(Zero Coupon Zero Principal)



NO INTEREST
(Zero Coupon)



NO PRINCIPAL RETURN
(Zero Principal)



Structured Donations.
Issued by NPOs on Social Stock Exchange (SSE)

KEY ACTORS

NPOs: Raise via ZCZPs, SIF donations, Development Bonds.
FPEs (For-Profit): Raise via Equity, Debt.

• 'INOPERATIVE' STATUS: AIFs with no remaining funds post-tenure can seek this status, reducing administrative compliance.

2. GLOBAL MARKETS: INVESTMENT NARRATIVES (2025-2026)

TINA: THERE IS NO ALTERNATIVE



Belief that US Equities (S&P 500) offer best returns.

Context: Strong US economy, energy shock resilience. Revived in 2026. (S&P 500 > 7,000)

TIARA: THERE IS A REAL ALTERNATIVE



Investors look beyond US to Europe & Emerging Markets (EMs) for valuations & diversification.

Shift: Weaker dollar made EMs attractive earlier. But recent outflows back to US suggest temporary eclipse of TIARA by TINA.

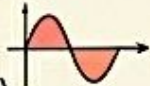
3. PHYSICS: INFRARED (IR) & QUANTUM SENSORS

Gamma Rays | X-Rays | Ultraviolet | **VISIBLE (Red)** | **INFRARED** | Microwaves | Radio Waves

Wavelength: ~700 nm to 1 mm

Nature of Infrared:

Transverse EM waves.
Invisible to naked eye, felt as Heat (Thermal Radiation).



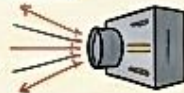
Quantum Interaction

- Wave-Particle Duality.
- Lower energy photons.
- Don't cause electronic transitions, but **DO** cause MOLECULAR VIBRATIONS (why absorbed as heat).



SENSOR SYSTEMS

Active Sensors:
Emit IR & measure reflection.
(e.g., LiDAR, TV remotes)



Passive Sensors:
Detect IR emitted by objects.
(e.g., Night-vision goggles, thermal imaging)



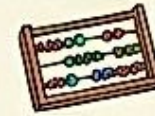
Energy

Quantum Sensors:
Utilize atomic quantum states to detect faint IR signatures. Crucial for **Deep-Space Astronomy, Stealth Detection.**

4. THE LEARNING CRISIS & FLN PUZZLE

LEARNING CRISIS:

Gap between 'Schooling' (Enrollment) & 'Learning' (Outcomes).



FLN: FOUNDATIONAL LITERACY & NUMERACY

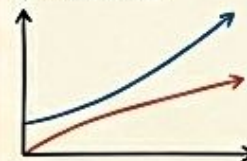
→ Ability to read with meaning & perform basic math by end of Grade 3.

THE PUZZLE OF SALIENCE

Prominence/Urgency a problem holds.



- Child's inability to read is **INVISIBLE** compared to a broken road.
- Leads to low salience: → Not prioritized by Parents or Politicians



SYSTEMIC ISSUES

Low-Learning Equilibrium: Everyone satisfied with enrollment, despite absent quality.

Power Asymmetry: Poor parents lack 'voice'; Middle class has 'exited' to private schools.



WAY FORWARD

NIPUN BHARAT MISSION:
Universal FLN by 2026-27.

→ **MAKE LEARNING VISIBLE:**
Through Community-led assessments. Increases political & social salience.